

Ryan Reece, Ph.D.

Machine learning engineer / data scientist / physicist

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EXPERIENCE

Sr. Staff Technical Lead and Manager, AI Cloud | Oct 2024 - present (1 yr 8 mo)

Staff Machine Learning Engineer | Oct 2022 - Oct 2024 (2 yr)

[Tenstorrent](#), Santa Clara, CA

- Co-led the AI Cloud team behind [console.tenstorrent.com](#), Tenstorrent's public demo platform showcasing record-breaking DeepSeek chat and Wan video generation on Galaxy hardware; unveiled at the TT-Deploy launch event on May 1, 2026 alongside the GA of Blackhole Galaxy servers
- Architected and maintain chat.tenstorrent.com, an internal chat service used company-wide for both interactive use and programmatic workloads (up to 450k requests/day from a single internal customer), proxying both Tenstorrent-hosted models and 3rd party APIs (Anthropic, Google, OpenAI)
- Drove the strategic decision to adopt vLLM as an inference server, recognizing that Continuous Batching and PagedAttention were essential for realistic multi-user chat; brought up Llama3 70B and later DeepSeek R1 on Tenstorrent hardware in Cloud through close collaboration with the TT Metal kernel team
- Developed inference API demos across language (BERT, T5, Llama, DeepSeek), image (YOLO), and video (Wan) models; benchmarked performance across multiple generations of Tenstorrent accelerators; launched Tenstorrent's NLP inference-as-a-service through EdenAI
- Worked with full-stack engineers to develop a cloud for push-button provisioning of Tenstorrent accelerators with Kubernetes; led containerization of the AI software stack
- Co-led design and delivery of a QLoRA fine-tuning service for customer-provided data: wrote the core fine-tuner (single GPU → multi-GPU), decomposed product requirements into 50+ tracked deliverables for my team, and drove the selection of Metaflow as the orchestrator for managing fine-tuning jobs on Kubernetes
- Collaborated with the ML training team to bring up Tenstorrent's first multi-host training examples, training a single model across multiple servers with combined data and model parallelism
- Led bringup of slurm orchestration for training job; helped design and implement slurm selector plugin for topologically-aware (really graph-aware) job scheduling
- Driving early experiments in agentic applications running on Tenstorrent hardware, backed by OpenClaw, to demonstrate Galaxy's suitability for agentic workloads beyond chat inference

Machine Learning Engineer | Apr 2018 - Aug 2022 (4 yrs 4 mos)

[Cerebras Systems](#), Sunnyvale, CA

- Developed end-to-end model references in both [pytorch](#) and [tensorflow](#), including the input data pipeline for Cerebras Wafer-Scale Engines
- Trained benchmark models and did exploratory optimization of various models for computer vision ([ResNets](#)) and NLP ([GNMT](#), [Transformer](#), [Linformer](#), [BERT](#), [RoBERTa](#), [GPT-2](#)); explored impacts of using [mixed precision](#), bucketing by sequence length, activation sparsity
- Model references and data pipeline code delivered to customers in the Model Zoo with several examples and detailed documentation
- Helped develop a new normalization layer, [OnlineNorm](#), that uses streaming statistics to allow normalization of activations with small batch sizes [[NeurIPS 2019](#)]
- Directly engaged and supported customers (from both national labs and industry) in meetings and on-sites; including helping Pittsburgh Supercomputing Center (PSC) select among scientific proposals for using their dual CS-1 systems: [Neocortex](#)
- Triaged, explored, and tested customer-shared models; represented customer requirements to compiler engineers; debugged model and data pipeline issues for customers
- Co-authored a blog about [Getting started with PyTorch BERT models on the Cerebras CS-2 System](#)

Postdoctoral Research Fellow | Jul 2013 - Aug 2017 (4 yrs 2 mos)

[Santa Cruz Institute for Particle Physics](#), The University of California, Santa Cruz, and
The European Council for Nuclear Research ([CERN](#)), Geneva, Switzerland

- 11 years (postdoc and Ph.D.) as a member of the ATLAS experiment, a 3000+ person collaboration looking for new physics in high energy proton-proton collisions at the Large Hadron Collider (LHC)
- Long involvement in codebase of more than 10 million lines of C++ and almost as many lines of Python
- [Expert in petabyte data reduction](#) (ATLAS ~10 PB/year), world-wide grid computing, and [data visualization](#) as a user and primary supporter of our group's 200-CPU computing cluster, accumulated more than [350k CPU-hours](#)
- Co-led analysis groups in multiple searches for signals of supersymmetry and exotic decays, contributed to 6 research publications, and defended their approval
- 2015-17, full-time support the operations of the [data acquisition system \(DAQ\)](#) and detector monitoring systems of the SCT (a tracking sub-detector in ATLAS)
- 2016-17, built more [expertise in machine learning techniques](#), deep learning frameworks using Keras to build CNNs for [particle classification](#), and another project using sklearn for anomaly detection by [clustering with Gaussian Mixture Models](#)

Graduate Researcher | Jun 2006 - Jul 2013 (7 yrs)

[The University of Pennsylvania](#), Philadelphia, PA, and
The European Council for Nuclear Research ([CERN](#)), Geneva, Switzerland

- First summers as a student with Penn (2006-08) at CERN participating in the [integration and commissioning of custom electronics](#) for the Transition Radiation Tracker (TRT), the outermost sub-detector of the ATLAS tracker
- 2009-12, throughout most of the running of the LHC, rotated the [on-call responsibility](#) for the TRT DAQ
- Ph.D. research with the data from ATLAS focused on the identification of decays of tau leptons and their use in searches for new physics, a [pattern recognition](#) problem to identify a type of particle
- 2009-10, was the lead developer of the cut-based tau identification used with the first ATLAS data
- 2010-12, helped develop advanced tau identification using [Boosted Decision Trees \(BDTs\)](#) which superseded the above
- Knack for developing data analysis frameworks: *e.g.* [pyframe](#) has been used by several analyses in ATLAS
- The ATLAS and CMS experiments at the LHC [discovered the long-sought-after Higgs boson](#), evidence of which was announced on July 4, 2012 [[Physics Letters B](#), [arxiv:1207.7214](#)]

EDUCATION

- **Ph.D. Experimental Particle Physics**, The University of Pennsylvania (Philadelphia, PA), Jun 2006 - Jul 2013
thesis: "[A search for new physics in high-mass ditau events in the ATLAS detector](#)"
- **B.S. Physics with Honors**, The University of Texas (Austin, TX), Aug 2003 - May 2006
thesis: "Late pulsing in the Hamamatsu R1408 PMT used in the Sudbury Neutrino Observatory"

PUBLICATIONS

- Chiley, V. *et al.* (2019). Online normalization for training neural networks. [NeurIPS 2019](#). [[arxiv:1905.05894](#)]
- Albertsson, K. *et al.* (2018). Machine learning in high energy physics community white paper. [[arxiv:1807.02876](#)]
- As a member of the ATLAS collaboration since June 1, 2008, I am an "author" of more than 800 publications ([google scholar](#), [inspire](#)). I made substantial contributions to [14 publications in scientific journals](#) and several more physics [conference proceedings](#). My favorites:
 1. Search for supersymmetry in a final state containing two photons and missing transverse momentum in $\sqrt{s} = 13$ TeV pp collisions at the LHC using the ATLAS detector. [European Physical Journal C](#), **76**, 517 (2016). [[arxiv:1606.09150](#)]
 2. Identification and energy calibration of hadronically decaying tau leptons with the ATLAS experiment in pp collisions at $\sqrt{s} = 8$ TeV. [European Physical Journal C](#), **75**, 303 (2015). [[arxiv:1412.7086](#)]

3. A search for high-mass resonances decaying to $\tau^+\tau^-$ with the ATLAS detector. *Physics Letters B*, 719, 242-260 (2013). [[arxiv:1210.6604](#)]
4. Performance of the ATLAS detector using first collision data. *Journal of High Energy Physics*, 9, 56 (2010). [[arxiv:1005.5254](#)]

SKILLS

- **General:** deep learning (NLP and CV), statistical analysis, data visualization, data-driven modeling, anomaly detection, neural network classifiers, boosted decision trees, petabyte data reduction, object-oriented design, polymorphic interfaces, writing technical reports, working independently and in groups, presenting my ideas, graduate level physics and mathematics
- **Programming languages, fluent:** C/C++/STL (23+ years), Python (20+ years);
experienced: Go, javascript, SQL; **Markup languages:** L^AT_EX, Markdown, html with css
- **ML / Data science software:** [pytorch](#), [tensorflow](#), [HuggingFace](#), [vLLM](#), LiteLLM, Metaflow, OpenClaw, keras, matplotlib, numpy, scipy, scikit-learn, pandas, jupyter, ROOT, pyhf, RooStats, TMVA
- **General software:** Linux, bash, git, slurm, kubernetes, AWS (EC2, S3), docker, Mathematica, QT, UML
- **Hobbies:** poker, philosophy, cycling, climbing

Last updated: May 22, 2026